

Recommender System (CS4053)

Final Exam

Date: 6/2/2025

Course Instructor

Syed Zain Ul Hassan

Total Time: 3hrs

Total Marks: 100

Total Questions: 05

Roll No

Section

Student Signature

Do not write below this line.

Attempt all questions.

CLO # 1: Describe different techniques in making personalized recommendations in various scenarios.

Q1: For each of the given scenarios and the choices provided, identify exactly one **most suitable** technique that should be used and provide a brief reason for your selection. **[Est. Time: 10 minutes, Points: 15]**

- i. An online pharmacy that allows users to purchase medicines upon uploading prescriptions.
 - a) User-based Collaborative Filtering
 - b) Content-based Filtering
 - c) Item-based Collaborative Filtering
 - d) Knowledge-graph based system
- ii. A store that allows users to book gold jewelry of various categories.
 - a) Content-based Filtering
 - b) Case-based system
 - c) Neural recommender system
 - d) Item-based Collaborative Filtering
- iii. An app that allows users to purchase latest laptops.
 - a) Probabilistic Collaborative Filtering
 - b) Case-based system
 - c) Neural recommender system
 - d) Transformer-driven system
- iv. A website that allows users to rate movies and read reviews.
 - a) Constraint-based system
 - b) Neural recommender system
 - c) Transformer-driven system
 - d) Item-based Collaborative Filtering
- v. An app that allows users to share photos and short videos.
 - a) Case-based system
 - b) Constraint-based system
 - c) User-based Collaborative Filtering
 - d) Content-based Filtering

CLO # 2 Solve mathematical optimization problems pertaining to recommender systems.

Q2 (a): Consider the data provided in **Table 1** of the data sheet. Use user-based Naïve Bayes collaborative filtering to find the missing rating $R(\text{item1}, \text{user2})$. Assume $\alpha = 0.01$. **[Est. Time: 25 minutes, Points: 10]**

Q2 (b): Do we need significance weighting when using any probabilistic collaborative filtering approach? Justify your answer with a proper reason. **[Est. Time: 10 minutes, Points: 10]**

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CLO # 1: Describe different techniques in making personalized recommendations in various scenarios.

Q3 (a): Consider an interaction table of size 60000 x 90000 on which Matrix Factorization technique is to be applied. Since the convergence is usually slow and non-deterministic, the following modification in the algorithm is proposed: **[Est. Time: 25 minutes, Points: 10]**

- i. After the first iteration of stochastic gradient descent, decompose each of the matrix factors into exactly two further matrices. We can call them *level2* factors.
- ii. Optimize each of the *level2* matrices using stochastic gradient descent.
- iii. After every 10 iterations, decompose each of the matrices into two further matrices. Such that after K decompositions they are called *levelK* factors until convergence is achieved, or at least one of the matrices cannot be decomposed any further.
- iv. Terminate the algorithm and report error.

Explain whether this updated algorithm will improve the performance and efficiency of the original Matrix Factorization technique. You are allowed to freely use any examples or diagrams to explain your answer.

Q3 (b): Consider the data in **Table 2** of the data sheet. Use content-based filtering to predict the web page to be recommended to the user. Assume **Page2** to be the active page. **[Est. Time: 20 minutes, Points: 10]**

CLO # 2 Solve mathematical optimization problems pertaining to recommender systems.

Q4 (a): Consider the data provided in **Table 3** of the data sheet. Use case-based approach to predict the item to be recommended to the user. **[Est. Time: 20 minutes, Points: 10]**

Q4 (b): Can a case-based system be affected by sparsity in interactions? Justify your answer with a proper reason. **[Est. Time: 15 minutes, Points: 10]**

CLO # 3 Discuss how a recommender system should be evaluated in terms of the performance and the user's satisfaction.

Q5 (a): In a typical neural recommendation approach, the user and item features are taken as input to predict the rating value as a class label. Consider the raw features provided in **Table 4** of the data sheet. If the neural recommender produces highly accurate results for this specific data, explain how it cannot produce reliable recommendations for any other users. **[Est. Time: 20 minutes, Points: 10]**

Q5 (b): Can a multi-label classifier be used to predict ratings for multiple items at once given user and item features? Justify your answer with a proper reason. **[Est. Time: 20 minutes, Points: 10]**

Q5 (c): Write Python code snippet to load data provided in **Table 1** to a dataframe named "*interactions*" and then print all the ratings given by *user 1*. **[Est. Time: 10 minutes, Points: 5]**

Good luck!